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<https://github.com/eic-r/cs587-fractional-residual>

- Goal:** Explore the potential benefits of incorporating fractional derivatives into neural network training

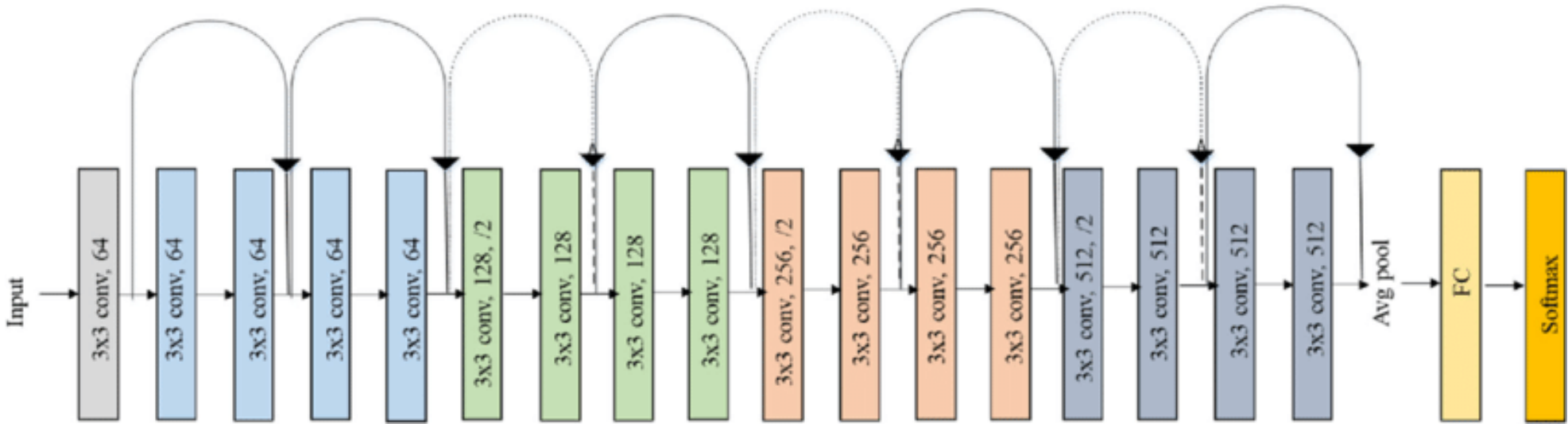
 - The potential of fractional derivatives for neural network training is currently underexplored.
 - This project incorporates fractional derivatives in a deep model.

Motivation

Past research has suggested that fractional derivatives can improve optimization dynamics in neural networks. Bao et al. (2018) implemented a deep neural network trained on the MNIST dataset using the Caputo derivative, testing various fractional orders. They found that performance peaked around an order of 10/9, slightly surpassing traditional integer-order derivatives, while also benefiting from L2 regularization to mitigate overfitting. Gomolka (2018) introduced a model utilizing the Grünwald-Letnikov derivative, demonstrating its ability to solve the XOR problem and confirming convergence.

Unlike standard gradients that depend only on the current state, fractional gradients blend information from past updates, potentially leading to better convergence, improved generalization, and reduced overfitting. However, prior studies mostly explored simple architectures and datasets.

This project investigates fractional backpropagation in a deeper network (ResNet-18) on CIFAR-10, aiming to assess whether fractional derivatives can offer consistent advantages in more complex learning tasks.

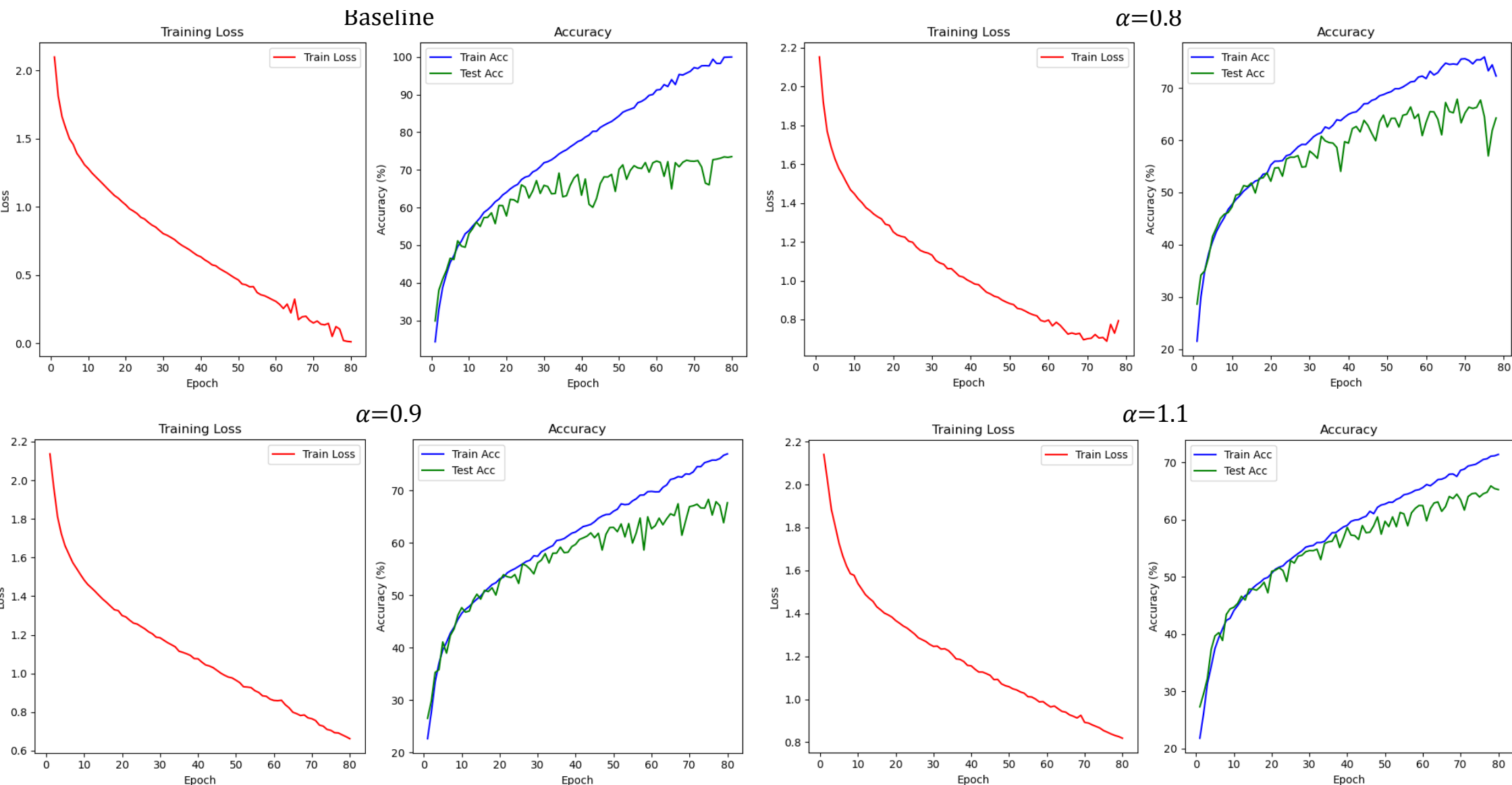


The original ResNet-18 architecture (Ramzan et al.). In the fractional model, the output of each colored group of blocks is wrapped so that during backpropagation its gradient is replaced with a fractional gradient.

Fractional Gradients

Idea: Fractional derivatives smooth gradient updates and blend past information.

$$D^\alpha f(t) = \lim_{h \rightarrow 0} \frac{1}{h^\alpha} \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} f(t - kh)$$



Model description

Architecture

A standard ResNet-18 architecture is modified to incorporate fractional derivatives into the gradient flow. Hooks are inserted after each residual block (layer1–layer4) to replace standard gradients with a Grünwald–Letnikov fractional derivative during backpropagation. Gradients are computed using a weighted history of past gradients, where the weights are determined by the generalized binomial coefficients based on a fractional order α . Models with $\alpha=0.7, 0.8, 0.9, 1.0, 1.1$ are trained with Adam optimizer with learning rate 0.001 and weight decay regularization and compared with a baseline ResNet-18 model.

Training objective

We train the model to minimize cross-entropy loss on CIFAR-10. During backpropagation, we replace standard gradients with fractional Grünwald–Letnikov derivatives.

Challenges

- Fractional model is sensitive to hyperparameters and difficult to tune
- The method is only an approximation of the Grünwald–Letnikov derivative as only a limited number of prior derivatives can be considered.

$$D^\alpha f(t) \approx \sum_{k=0}^N (-1)^k \binom{\alpha}{k} f(t - kh)$$

Experiments

- Dataset:** CIFAR-10 is an image classification dataset consisting of 60,000 color images (32×32 pixels) divided into 10 classes.
- An alpha of 0.9 performed the best out of the fractional models after 80 epochs, but could not beat baseline ResNet-18.
- The difference between test and training accuracy is slightly smaller, possibly indicating improved generalization.
- Fractional models are more complex and require significantly more time to train.

	Base	0.7 (40 epochs)	0.8	0.9	1.0	1.1
Loss	0.013	1.028	0.707	0.663	0.739	0.819
Train Acc	99.98%	63.76%	75.38%	76.97%	74.47%	71.43%
Test Acc	73.50%	61.21%	67.68%	67.65%	67.59%	65.28%
Time (s)	11832.60	N/A	14400	14086.76	7038.91	11767.64